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Mahindra University École C
Hyde
Minor II E
Program: B. Tec
Year: I Semester: I
Subject: Calculus and ODE (MA 1101)

Date: 29-10-25 Time: 10 a.m.
Time Duration: 90 Minutes Max. Marks: 20

Instructions:

1. There are 4 questions, all of which are compulsory.
2. The order of answers should be same as the order of questions.
3. Justify your answer wherever required. Guesswork will not be considered in evaluation.

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1. Without attempting to evaluate the integral, compute $f'(x)$ if f is defined by the formula $f(x) = \int_{x^3}^{x^2} \frac{2t^{10}}{1+t^4}\,dt$. M1

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2. Determine if the following series converges or diverges. If it converges find its value: M

$$\sum_{n=1}^{\infty} \frac{1}{n^2+4n+3}.$$

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3. Solve the ordinary differential equation

$$y' + \frac{2}{x}y = x^2y^3, \qquad x > 0.$$

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4. Evaluate the following limit if it exists, or show that it does not exist: M

$$\lim_{(x,y)\to(0,0)} \frac{x^n y}{x^{2n}+y^2},$$

where $n \geq 2$ is an integer.

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